

$$\left\{ \begin{array}{ll} \frac{\partial f}{\partial t} = \frac{\partial}{\partial s} \left(\frac{a}{f^2} \frac{\partial f}{\partial s} \right) & s \in (0, 82), t > 0 \\ \frac{a}{f(0)^2} \frac{\partial f}{\partial s} \Big|_{f=0} = 0 & s=0, t > 0 \\ \frac{a}{f(82)^2} \frac{\partial f}{\partial s} \Big|_{f=82} = 0 & s=82, t > 0 \\ f(s, 0) = f(s) & s \in (0, 82), t=0 \end{array} \right.$$

C is parameterized by $\vec{r}(z) = \langle x(z), 0, z \rangle$

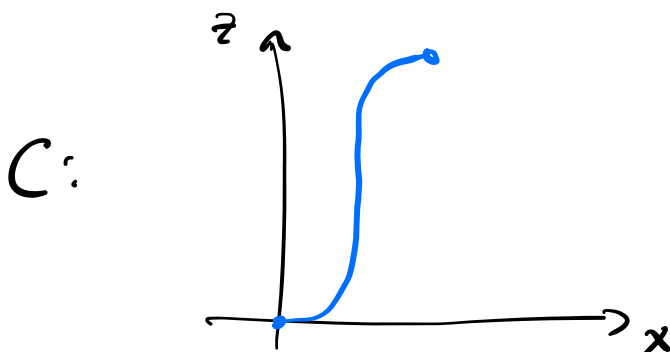
where $s=0 \iff \vec{r}(0) = \langle 0, 0, 0 \rangle$

$s=82 \iff \vec{r}(78) = \langle x(78), 0, 78 \rangle$

Here, $x(z) = r_b(1 - e^{-az}) + r_t e^{-a(z-78)}$.

$$\frac{ds}{dz} = |\vec{r}'(z)| = \sqrt{x'(z)^2 + 0^2 + 1^2}$$

$$\Rightarrow \boxed{ds = \sqrt{x'(z)^2 + 1} dz}$$



$$\left\{ \begin{array}{l} \frac{\partial q}{\partial t} = \frac{1}{\sqrt{x(z)^2+1}} \frac{\partial}{\partial z} \left(\frac{\alpha}{f^2} \cdot \frac{1}{\sqrt{x(z)^2+1}} \frac{\partial q}{\partial z} \right) \quad z \in (0, 78), t > 0 \\ \frac{1}{\sqrt{x(z)^2+1}} \frac{\alpha}{f(z)^2} \frac{\partial q}{\partial z} \Big|_{z=0} = 0 \quad z=0, t > 0 \\ q(78, t) = 1 \quad z=78, t > 0 \\ q(z, 0) = f(z) \quad t=0, z \in (0, 78) \end{array} \right.$$

$$D_z = \frac{1}{dz} \begin{bmatrix} -3/2 & 2 & -1/2 & 0 & \dots & \dots & \dots & 0 \\ -1/2 & 0 & 1/2 & 0 & \dots & \dots & \dots & 0 \\ 0 & -1/2 & 0 & 1/2 & \dots & \dots & \dots & 0 \\ & & \ddots & \ddots & \ddots & \ddots & \ddots & \\ 0 & \dots & \dots & \dots & 0 & -1/2 & 0 & 1/2 \\ 0 & \dots & \dots & \dots & 0 & 1/2 & -2 & 3/2 \end{bmatrix}$$

when $i=0$: $\frac{\partial q}{\partial z} = 0 \Rightarrow \frac{-\frac{3}{2}q_0 + 2q_1 - \frac{1}{2}q_2}{dz} = 0$

$$\Rightarrow \boxed{q_0 = \frac{-2q_1 + \frac{1}{2}q_2}{-3/2}}$$

when $i=N_z$: $\boxed{q_{N_z} = 1}$

function $dqdt = \text{RHS_StripStrip}(t, q, Dz)$

$$q-L = \frac{-2q(2) + 1/2 q(3)}{-3/2}$$

$$q-R = 1$$

$$q_{\text{full}} = [q-L; q; q-R]$$

$$dqdt = \frac{1}{\sqrt{x(z)^2 + 1}} .* \left(Dz .* \left(\frac{1}{\sqrt{x(z)^2 + 1}} .* \frac{\alpha}{q_{\text{full}}^2} .* (Dz .* q_{\text{full}}) \right) \right)$$

$$dqdt = dqdt(2:end-1);$$